

Chapter V

Fractal Dimension

1 Fractal Dimension

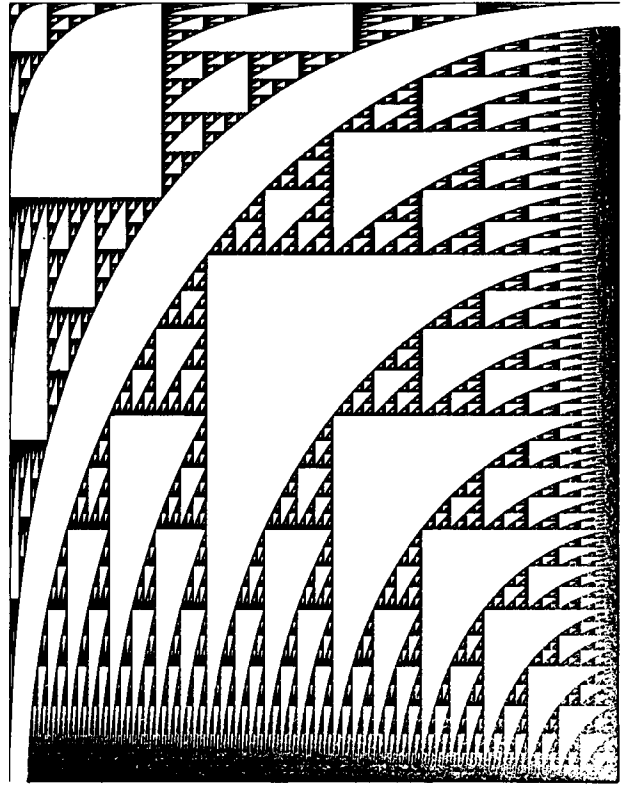
How big is a fractal? When are two fractals similar to one another in some sense? What experimental measurements might we make to tell if two different fractals may be metrically equivalent? What is the same about the two fractals in Figure V.131?

There are various numbers, associated with fractals, which can be used to compare them. They are generally referred to as fractal dimensions. They are attempts to quantify a subjective feeling we have about how densely the fractal occupies the metric space in which it lies. Fractal dimensions provide an objective means for comparing fractals.

Fractal dimensions are important because they can be defined in connection with real-world data, and they can be measured approximately by means of experiments. For example, one can measure the “fractal dimension” of the coastline of Great Britain; its value is about 1.2. Fractal dimensions can be attached to clouds, trees, coastlines, feathers, networks of neurons in the body, dust in the air at an instant in time, the clothes you are wearing, the distribution of frequencies of light reflected by a flower, the colors emitted by the sun, and the wrinkled surface of the sea during a storm. These numbers allow us to compare sets in the real world with the laboratory fractals, such as attractors of IFS.

We restrict attention to compact subsets of metric spaces. This fits well with the idea of modelling the real physical world by subsets of metric spaces. Suppose that an experimentalist is studying a physical entity, and he wishes to model this entity by means of a subset of $\mathbb{R}^3$. Then he can use a compact set for his model. For example, he can assume that the distances he measures are Euclidean distances, and he can assume that the universe is bounded. He can assume that any Cauchy sequence of points in his model set converges to a point in his model set, because he cannot experimentally invalidate this assumption. Although mathematically we can distinguish between a set and its closure, we cannot make the same distinction

Figure V.131. Do the two implied fractals have the same dimension?



between their physical counterparts. The assumption of compactness will allow the model to be handled theoretically with relative ease.

Let (X, d) denote a complete metric space. Let $A \in \mathcal{H}(X)$ be a nonempty compact subset of X . Let $\epsilon > 0$. Let $B(x, \epsilon)$ denote the closed ball of radius ϵ and center at a point $x \in X$. We wish to define an integer, $\mathcal{N}(A, \epsilon)$, to be the least number of closed balls of radius ϵ needed to cover the set A . That is

$$\mathcal{N}(A, \epsilon) = \text{smallest positive integer } M \text{ such that } A \subset \bigcup_{n=1}^M B(x_n, \epsilon),$$

for some set of distinct points $\{x_n : n = 1, 2, \dots, M\} \subset X$. How do we know that there is such a number $\mathcal{N}(A, \epsilon)$? Easy! The logic is this: surround every point $x \in A$ by an *open* ball of radius $\epsilon > 0$ to provide a cover of A by open sets. Because A is compact this cover possesses a finite subcover, consisting of an integer number, say $\hat{M}$, of open balls. By taking the closure of each ball, we obtain a cover consisting of $\hat{M}$ *closed* balls. Let C denote the set of covers of A by at most $\hat{M}$ closed balls of radius ϵ . Then C contains at least one element. Let $f : C \rightarrow \{1, 2, 3, \dots, \hat{M}\}$ be defined by $f(c) = \text{number of balls in the cover } c \in C$. Then $\{f(c) : c \in C\}$ is a finite set of positive integers. It follows that it contains a least integer, $\mathcal{N}(A, \epsilon)$.

The intuitive idea behind fractal dimension is that a set A has fractal dimension D if:

$\mathcal{N}(A, \epsilon) \approx C\epsilon^{-D}$ for some positive constant C .

Here we use the notation “ $\approx$ ” as follows. Let $f(\epsilon)$ and $g(\epsilon)$ be real valued functions of the positive real variable ϵ . Then $f(\epsilon) \approx g(\epsilon)$ means that $\lim_{\epsilon \rightarrow 0} \{\ln(f(\epsilon))/\ln(g(\epsilon))\} = 1$. If we “solve” for D we find that

$$D \approx \frac{\ln \mathcal{N}(A, \epsilon) - \ln C}{\ln(1/\epsilon)}.$$

We use the notation $\ln(x)$ to denote the logarithm to the base e of the positive real number x . Now notice that the term $\ln C / \ln(1/\epsilon)$ approaches 0 as $\epsilon \rightarrow 0$. This leads us to the following definition.

Definition 1.1 Let $A \in \mathcal{H}(X)$ where (X, d) is a metric space. For each $\epsilon > 0$ let $\mathcal{N}(A, \epsilon)$ denote the smallest number of closed balls of radius $\epsilon > 0$ needed to cover A . If

$$D = \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\}$$

exists, then D is called the fractal dimension of A . We will also use the notation $D = D(A)$ and will say “ A has fractal dimension D .”

Examples & Exercises

1.1. This example takes place in the metric space $(\mathbb{R}^2, \text{Euclidean})$. Let $a \in X$ and let $A = \{a\}$. A consists of a single point in the space. For each $\epsilon > 0$, $\mathcal{N}(A, \epsilon) = 1$. It follows that $D(A) = 0$.

1.2. This example takes place in the metric space $(\mathbb{R}^2, \text{Manhattan})$. Let A denote the line segment $[0, 1]$. Let $\epsilon > 0$. Then it is quite easy to see that $\mathcal{N}(A, \epsilon) = \lceil -[-1/\epsilon] \rceil$, where $\lceil x \rceil$ denotes the integer part of the real number x . In Figure V.132 we have plotted the graph of $\ln(\mathcal{N}(A, \epsilon))$ as a function of $\ln(1/\epsilon)$. Despite a rough start, it appears clear that

$$\lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\} = 1.$$

In fact for $0 < \epsilon < 1$

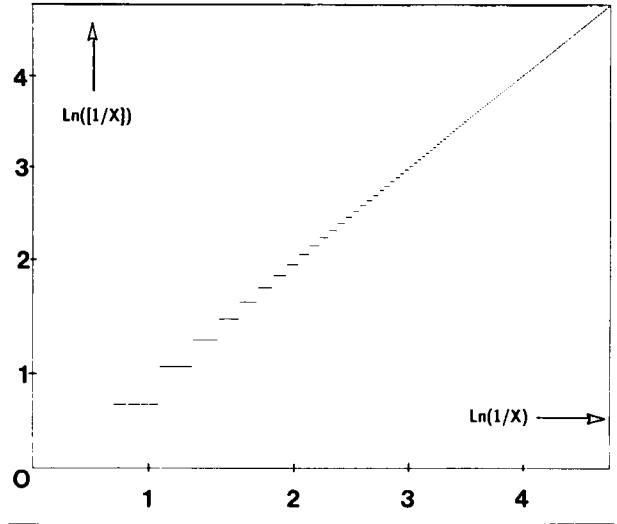
$$\frac{\ln(1/\epsilon)}{\ln(1/\epsilon)} \leq \frac{\ln(-[-1/\epsilon])}{\ln(1/\epsilon)} = \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \leq \frac{\ln(1/\epsilon + 1)}{\ln(1/\epsilon)} = \frac{\ln(1 + \epsilon) + \ln(1/\epsilon)}{\ln(1/\epsilon)},$$

Both sides here converge to 1 as $\epsilon \rightarrow 0$. Hence the quantity in the middle also converges to 1. We conclude that the fractal dimension of a closed line segment is one. We would have obtained the same result if we had used the Euclidean metric.

1.3. Let (X, d) be a metric space. Let $a, b, c \in X$, and let $A = \{a, b, c\}$. Prove that $D(A) = 0$.

The following two theorems simplify the process of calculating the fractal dimension. They allow one to replace the continuous variable ϵ by a discrete variable.

Figure V.132. Plot of $\ln([1/x])$ as a function of $\ln(1/x)$. This illustrates that in the computation of the fractal dimension one usually evaluates the limiting “slope” of a discontinuous function. In the present example this slope is 1.



Theorem 1.1 Let $A \in \mathcal{H}(X)$, where (X, d) is a metric space. Let $\epsilon_n = Cr^n$ for real numbers $0 < r < 1$ and $C > 0$, and integers $n = 1, 2, 3, \dots$. If

$$D = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_n))}{\ln(1/\epsilon_n)} \right\},$$

then A has fractal dimension D .

Proof Let the real numbers r and C , and the sequence of numbers $E = \{\epsilon_n : n = 1, 2, 3, \dots\}$ be as defined in the statement of the theorem. Define $f(\epsilon) = \max\{\epsilon_n \in E : \epsilon_n \leq \epsilon\}$. Assume that $\epsilon \leq r$. Then

$$f(\epsilon) \leq \epsilon \leq f(\epsilon)/r \text{ and } \mathcal{N}(A, f(\epsilon)) \geq \mathcal{N}(A, \epsilon) \geq \mathcal{N}(A, f(\epsilon)/r).$$

Since $\ln(x)$ is an increasing positive function of x for $x \geq 1$, it follows that

$$\left\{ \frac{\ln(\mathcal{N}(A, f(\epsilon)/r))}{\ln(1/f(\epsilon))} \right\} \leq \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\} \quad (1)$$

$$\leq \left\{ \frac{\ln(\mathcal{N}(A, f(\epsilon)))}{\ln(r/f(\epsilon))} \right\}. \quad (2)$$

Assume that $\mathcal{N}(A; \epsilon) \rightarrow \infty$ as $\epsilon \rightarrow 0$; if not then the theorem is true. The right-hand side of equation 2 obeys

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, f(\epsilon)))}{\ln(r/f(\epsilon))} \right\} &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_n))}{\ln(r/\epsilon_n)} \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_n))}{\ln(r) + \ln(1/\epsilon_n)} \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_n))}{\ln(1/\epsilon_n)} \right\}. \end{aligned}$$

The left-hand side of equation 2 obeys

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, f(\epsilon)/r))}{\ln(1/f(\epsilon))} \right\} &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_{n-1}))}{\ln(1/\epsilon_n)} \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_{n-1}))}{\ln(1/r) + \ln(1/\epsilon_{n-1}))} \right\} \\ &= \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon_n))}{\ln(1/\epsilon_n)} \right\}. \end{aligned}$$

So as $\epsilon \rightarrow 0$ both the left-hand side and the right-hand side of equation 2 approach the same value, claimed in the theorem. By the Sandwich Theorem of calculus, the limit as $\epsilon \rightarrow 0$ of the quantity in the middle of equation 2 also exists, and it equals the same value. This completes the proof of the theorem.

Theorem 1.2 The Box Counting Theorem. *Let $A \in \mathcal{H}(\mathbb{R}^m)$, where the Euclidean metric is used. Cover $\mathbb{R}^m$ by closed square boxes of side length $(1/2^n)$, as exemplified in Figure V.133 for $n = 2$ and $m = 2$. Let $\mathcal{N}_n(A)$ denote the number of boxes of side length $(1/2^n)$ which intersect the attractor. If*

$$D = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}_n(A))}{\ln(2^n)} \right\},$$

then A has fractal dimension D .

Proof We observe that for $m = 1, 2, 3, \dots$,

$$2^{-m} \mathcal{N}_{n-1} \leq \mathcal{N}(A, 1/2^n) \leq \mathcal{N}_{k(n)} \text{ for all } n = 1, 2, 3, \dots,$$

where $k(n)$ is the smallest integer k satisfying $k \geq n - 1 + \frac{1}{2} \log_2 m$. The first inequality holds because a ball of radius $1/2^n$ can intersect at most 2^m “on-grid” boxes of side $1/2^{n-1}$. The second follows from the fact that a box of side s can fit inside a ball of radius r provided $r^2 \geq (\frac{s}{2})^2 + (\frac{s}{2})^2 + \dots + (\frac{s}{2})^2 = m(\frac{s}{2})^2$ by the theorem of Pythagoras. Now

$$\lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}_{k(n)})}{\ln(2^n)} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(2^{k(n)})}{\ln(2^n)} \frac{\ln(\mathcal{N}_{k(n)})}{\ln(2^{k(n)})} \right\} = D,$$

since $\frac{k(n)}{n} \rightarrow 1$. Since also

$$\lim_{n \rightarrow \infty} \left\{ \frac{\ln 2^{-m} \mathcal{N}_{n-1}}{\ln(2^n)} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{\ln \mathcal{N}_{n-1}}{\ln(2^{n-1})} \right\} = D,$$

Theorem 1.1 with $r = 1/2$ completes the proof.

There is nothing magical about using boxes of side $(1/2)^n$ in Theorem 1.2. One can equally well use boxes of side Cr^n , where $C > 0$ and $0 < r < 1$ are fixed real numbers.

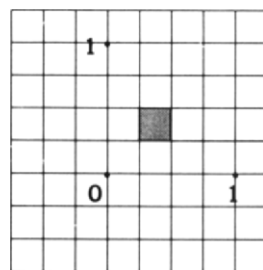
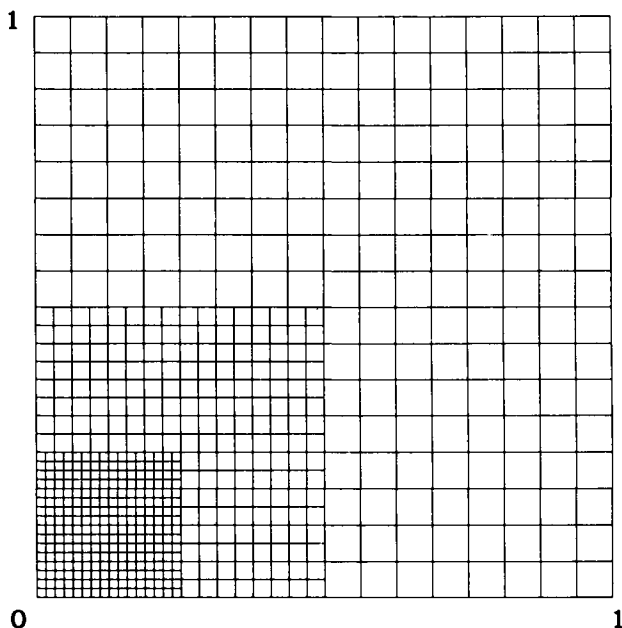


Figure V.133. Closed boxes of side $(1/2^n)$ cover $\mathbb{R}^2$. Here $n = 2$. See Theorem 1.2.

Figure V.134. It requires $(1/2^n)^{-2}$ boxes of side $(1/2^n)$ to cover $\blacksquare \subset \mathbb{R}^2$. We deduce, with a feeling of relief, that the fractal dimension of $\blacksquare$ is 2. To which collage is this related?



Examples & Exercises

1.4. Consider the $\blacksquare \subset \mathbb{R}^2$. It is easy to see that $\mathcal{N}_1(\blacksquare) = 4$, $\mathcal{N}_2(\blacksquare) = 16$, $\mathcal{N}_3(\blacksquare) = 64$, $\mathcal{N}_4(\blacksquare) = 256$, and in general that $\mathcal{N}_n(\blacksquare) = 4^n$ for $n = 1, 2, 3, \dots$; see Figure V.134.

Theorem 1.2 implies that

$$D(\blacksquare) = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}_n(\blacksquare))}{\ln(2^n)} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(4^n)}{\ln(2^n)} \right\} = 2.$$

1.5. Consider the Sierpinski triangle Δ , in Figure V.135, as a subset of $(\mathbb{R}^2, \text{Euclidean})$.

We see that $\mathcal{N}_1(\Delta) = 3$, $\mathcal{N}_2(\Delta) = 9$, $\mathcal{N}_3(\Delta) = 27$, $\mathcal{N}_4(\Delta) = 81$, and in general $\mathcal{N}_n(\Delta) = 3^n$ for $n = 1, 2, 3, \dots$

Theorem 1.2 implies that

$$D(\Delta) = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(\mathcal{N}_n(\Delta))}{\ln(2^n)} \right\} = \lim_{n \rightarrow \infty} \left\{ \frac{\ln(3^n)}{\ln(2^n)} \right\} = \frac{\ln(3)}{\ln(2)}.$$

1.6. Use the Box Counting Theorem, but with boxes of side length $(1/3)^n$, to calculate the fractal dimension of the classical Cantor set $\mathcal{C}$ described in exercise 1.5 in Chapter III.

1.7. Use the Box Counting Theorem to estimate the fractal dimension of the fractal subset of $\mathbb{R}^2$ shown in Figure V.136. You will need to take as your first box the

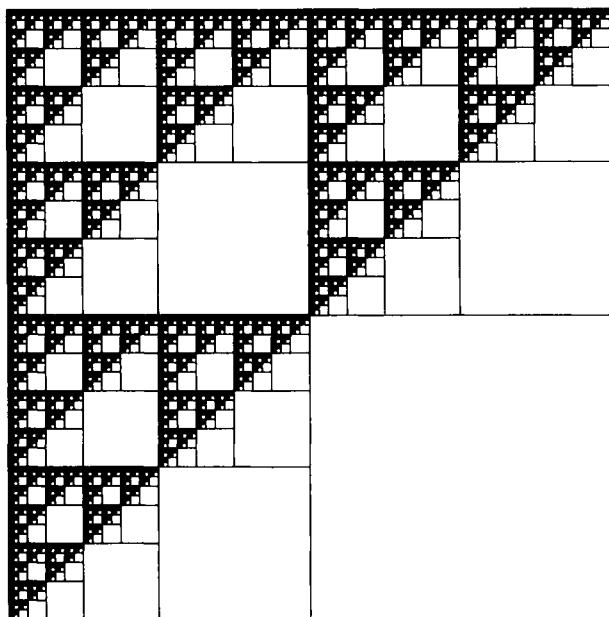


Figure V.135. It requires 3^n closed boxes of side $(1/2)^n$ to cover the Sierpinski triangle $\Delta \subset \mathbb{R}^2$. We deduce that its fractal dimension is $\ln(3)/\ln(2)$.

obvious one suggested by the figure. You should then find that there appears to be a pattern to the sequence of numbers $\mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_3, \dots$

1.8. The same problem as 1.7, this time applied to Figure V.137. By making the right choice of Cartesian coordinate system, you will make this problem easy.

What happens to the fractal dimension of a set if we deform it “with bounded distortion”? The following theorem tells us that metrically equivalent sets have the same fractal dimension. For example, the two fractals in Figure V.131 have the same fractal dimension!

Theorem 1.3 *Let the metric spaces (X_1, d_1) and (X_2, d_2) be metrically equivalent. Let $\theta : X_1 \rightarrow X_2$ be a transformation that provides the equivalence of the spaces. Let $A_1 \in \mathcal{H}(X_1)$ have fractal dimension D . Then $A_2 = \theta(A_1)$ has fractal dimension D . That is*

$$D(A_1) = D(\theta(A_2)).$$

Proof This proof makes use of the concepts of the lim sup and lim inf of a function. (The lim sup is discussed briefly following Definition 2.1 in the next section.)

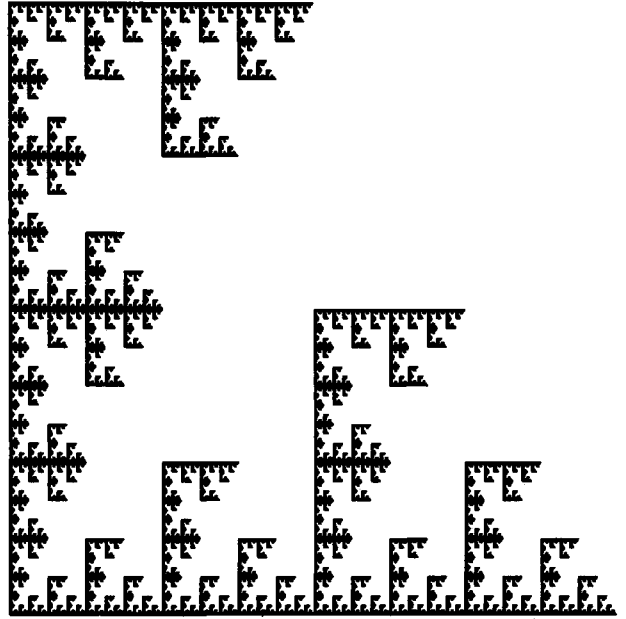
Since the two spaces (X_1, d_1) and (X_2, d_2) are equivalent under θ , there exist positive constants e_1 and e_2 such that

$$e_1 d_2(\theta(x), \theta(y)) < d_1(x, y) < e_2 d_2(\theta(x), \theta(y)) \quad \text{for all } x, y \in X_1. \quad (3)$$

Without loss of generality we assume that $e_1 < 1 < e_2$. Equation 3 implies

$$d_2(\theta(x), \theta(y)) < \frac{d_1(x, y)}{e_1} \quad \text{for all } x, y \in X_1.$$

Figure V.136. What other well-known fractal has the same fractal dimension?



This implies

$$\theta(B(x, \epsilon)) \subset B(\theta(x), \epsilon/e_1) \quad \text{for all } x \in X_1. \quad (4)$$

Now, from the definition of $\mathcal{N}(A_1, \epsilon)$, we know that there is a set of points $\{x_1, x_2, \dots, x_N\} \subset X_1$, where $N = \mathcal{N}(A_1, \epsilon)$, such that the set of closed balls $\{B(x_n, \epsilon) : n = 1, 2, \dots, N(A_1, \epsilon)\}$ provides a cover of A_1 . It follows that $\{\theta(B(x_n, \epsilon)) : n = 1, 2, \dots, N(A_1, \epsilon)\}$ provides a cover of A_2 . Equation 4 now implies that $\{B(\theta(x_n), \epsilon/e_1) : n = 1, 2, \dots, N(A_1, \epsilon)\}$ provides a cover of A_2 . Hence

$$\mathcal{N}(A_2, \epsilon/e_1) \leq \mathcal{N}(A_1, \epsilon).$$

Hence, when $\epsilon < 1$,

$$\frac{\ln(\mathcal{N}(A_2, \epsilon/e_1))}{\ln(1/\epsilon)} \leq \frac{\ln(\mathcal{N}(A_1, \epsilon))}{\ln(1/\epsilon)}.$$

It follows that

$$\limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)} \right\} \quad (5)$$

$$= \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon/e_1))}{\ln(1/\epsilon)} \right\} \leq \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_1, \epsilon))}{\ln(1/\epsilon)} \right\} = D(A_1). \quad (6)$$

We now seek an inequality in the opposite direction. Equation 3 implies that

$$d_1(\theta^{-1}(x), \theta^{-1}(y)) < e_2 d_2(x, y) \quad \text{for all } x, y \in X_2.$$

This tells us that

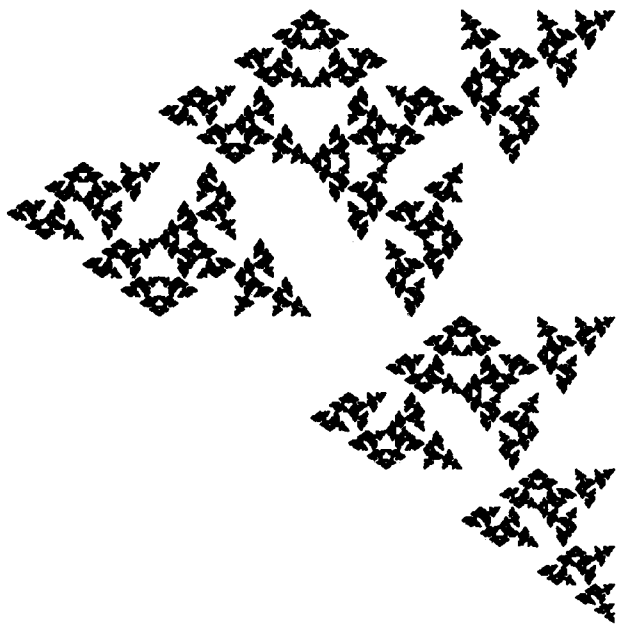


Figure V.137. If you choose the “first” box just right, the fractal dimension of this fractal is easily estimated. Count the number $\mathcal{N}_n$ of boxes of side $1/2^n$ which intersect the set, for $n = 1, 2, 3, \dots$, and apply the Box Counting Theorem.

$$\theta^{-1}(B(x, \epsilon)) \subset B(\theta^{-1}(x), e_2\epsilon)) \text{ for all } x \in X_2,$$

and this in turn implies

$$\mathcal{N}(A_1, e_2\epsilon) \leq \mathcal{N}(A_2, \epsilon).$$

Hence, when $\epsilon < 1$,

$$\frac{\ln(\mathcal{N}(A_1, e_2\epsilon))}{\ln(1/\epsilon)} \leq \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)}.$$

It follows that

$$\begin{aligned} D(A_1) &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_1, \epsilon))}{\ln(1/\epsilon)} \right\} \\ &= \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_1, e_2\epsilon))}{\ln(1/\epsilon)} \right\} \\ &\leq \liminf_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)} \right\}. \end{aligned} \tag{7}$$

By combining equations 6 and 7 we obtain

$$\liminf_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)} \right\} = D(A_1) = \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)} \right\}.$$

From this it follows that

$$D(A_2) = \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A_2, \epsilon))}{\ln(1/\epsilon)} \right\} = D(A_1).$$

This completes the proof.

Examples & Exercises

1.9. Let $\mathcal{C}$ denote the classical Cantor set, living in $[0, 1]$ and obtained by omitting “middle thirds.” Let $\tilde{\mathcal{C}}$ denote the Cantor set obtained by starting from the closed interval $[0, 3]$ and omitting “middle thirds.” Use Theorem 1.3 to show that they have the same fractal dimension. Verify the conclusion by means of a box-counting argument.

1.10. Let A be a compact nonempty subset of $\mathbb{R}^2$. Suppose that A has fractal dimension D_1 when evaluated using the Euclidean metric and fractal dimension D_2 when evaluated using the Manhattan metric. Show that $D_1 = D_2$.

1.11. This example takes place in the metric space $(\mathbb{R}^2, \text{Manhattan})$. Let A_1 and A_2 denote the attractors of the following two hyperbolic IFS

$$\{\mathbb{R}^2; w_1(x, y), w_2(x, y), w_3(x, y)\} \quad \text{and} \quad \{\mathbb{R}^2; w_4(x, y), w_5(x, y), w_6(x, y)\},$$

where

$$w_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$w_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

$$w_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and

$$w_4 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix},$$

$$w_5 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

$$w_6 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

By finding a suitable change of coordinates, show A_1 and A_2 have the same fractal dimensions.

2 The Theoretical Determination of the Fractal Dimension

The following definition extends Definition 1.1. It provides a value for the fractal dimension for a wider collection of sets.

Definition 2.1 Let (X, d) be a complete metric space. Let $A \in \mathcal{H}(X)$. Let $\mathcal{N}(\epsilon)$ denote the minimum number of balls of radius ϵ needed to cover A . If

$$D = \lim_{\epsilon \rightarrow 0} \left\{ \sup \left\{ \frac{\ln \mathcal{N}(\tilde{\epsilon})}{\ln(1/\tilde{\epsilon})} : \tilde{\epsilon} \in (0, \epsilon) \right\} \right\}$$

exists, then D is called the fractal dimension of A . We will also use the notation $D = D(A)$, and will say “ A has fractal dimension D .”

In stating this definition we have “spelled out” the $\lim \sup$. For any function $f(\epsilon)$, defined for $0 < \epsilon < 1$, for example, we have

$$\lim_{\epsilon \rightarrow 0} \sup f(\epsilon) = \lim_{\epsilon \rightarrow 0} \{\sup\{f(\epsilon) : \tilde{\epsilon} \in (0, \epsilon)\}\}.$$

It can be proved that Definition 2.1 is consistent with Definition 1.1: if a set has fractal dimension D according to Definition 1.1 then it has the same dimension according to Definition 2.1. Also, all of the theorems in this book apply with either definition. The broader definition provides a fractal dimension in some cases where the previous definition makes no assertion.

Theorem 2.1 *Let m be a positive integer; and consider the metric space $(\mathbb{R}^m, \text{Euclidean})$. The fractal dimension $D(A)$ exists for all $A \in \mathcal{H}(\mathbb{R}^m)$. Let $B \in \mathcal{H}(\mathbb{R}^m)$ be such that $A \subset B$; and let $D(B)$ denote the fractal dimension of B . Then $D(A) \leq D(B)$. In particular,*

$$0 \leq D(A) \leq m.$$

Proof We prove the theorem for the case $m = 2$. Without loss of generality we can suppose that $A \subset \blacksquare$. It follows that $\mathcal{N}(A, \epsilon) \leq \mathcal{N}(\blacksquare, \epsilon)$ for all $\epsilon > 0$. Hence for all ϵ such that $0 < \epsilon < 1$ we have

$$0 \leq \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \leq \frac{\ln(\mathcal{N}(\blacksquare, \epsilon))}{\ln(1/\epsilon)}.$$

It follows that

$$\lim_{\epsilon \rightarrow 0} \sup \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\} \leq \lim_{\epsilon \rightarrow 0} \sup \left\{ \frac{\ln(\mathcal{N}(\blacksquare, \epsilon))}{\ln(1/\epsilon)} \right\}.$$

The $\lim \sup$ on the right-hand side exists and has value 2. It follows that the $\lim \sup$ on the left-hand side exists and is bounded above by 2. Hence the fractal dimension $D(A)$ is defined and bounded above by 2. Also $D(A)$ is nonnegative.

If $A, B \in \mathcal{H}(\mathbb{R}^2)$ with $A \subset B$, then the fractal dimensions of A and B are defined. The above argument wherein $\blacksquare$ is replaced by B shows that $D(A) \leq D(B)$. This completes the proof.

The following theorem helps us to calculate the fractal dimension of the union of two sets.

Theorem 2.2 *Let m be a positive integer; and consider the metric space $(\mathbb{R}^m, \text{Euclidean})$. Let A and B belong to $\mathcal{H}(\mathbb{R}^m)$. Let A be such that its fractal dimension is given by*

$$D(A) = \lim_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\}.$$

Let $D(B)$ and $D(A \cup B)$ denote the fractal dimensions of B and $A \cup B$, respectively. Suppose that $D(B) < D(A)$. Then

$$D(A \cup B) = D(A).$$

Proof From Theorem 2.1 it follows that $D(A \cup B) \geq D(A)$. We want to show that $D(A \cup B) \leq D(A)$. We begin by observing that, for all $\epsilon > 0$,

$$\mathcal{N}(A \cup B, \epsilon) \leq \mathcal{N}(A, \epsilon) + \mathcal{N}(B, \epsilon).$$

It follows that

$$\begin{aligned} D(A \cup B) &= \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A \cup B, \epsilon))}{\ln(1/\epsilon)} \right\} \\ &\leq \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon) + \mathcal{N}(B, \epsilon))}{\ln(1/\epsilon)} \right\} \\ &\leq \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\} + \limsup_{\epsilon \rightarrow 0} \left\{ \frac{\ln(1 + \mathcal{N}(B, \epsilon)/\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \right\}. \end{aligned}$$

The proof is completed by showing that $\mathcal{N}(B, \epsilon)/\mathcal{N}(A, \epsilon)$ is less than 1 when ϵ is sufficiently small. This would imply that the second limit on the right here is equal to zero. The first limit on the right converges to $D(A)$.

Notice that

$$\sup \left\{ \frac{\ln(\mathcal{N}(B, \tilde{\epsilon}))}{\ln(1/\tilde{\epsilon})} : \tilde{\epsilon} < \epsilon \right\}$$

is a decreasing function of the positive variable ϵ . It follows that

$$\frac{\ln(\mathcal{N}(B, \epsilon))}{\ln(1/\epsilon)} < D(A) \text{ for all sufficiently small } \epsilon > 0.$$

Because the limit explicitly stated in the theorem exists, it follows that

$$\frac{\ln(\mathcal{N}(B, \epsilon))}{\ln(1/\epsilon)} < \frac{\ln(\mathcal{N}(A, \epsilon))}{\ln(1/\epsilon)} \text{ for all sufficiently small } \epsilon > 0.$$

This allows us to conclude that

$$\frac{\mathcal{N}(B, \epsilon)}{\mathcal{N}(A, \epsilon)} < 1 \text{ for all sufficiently small } \epsilon > 0.$$

This completes the proof.

Examples & Exercises

2.1. The fractal dimension of the hairy set $A \subset \mathbb{R}^2$, suggested in Figure V.138, is 2. The contribution from the hairs to $\mathcal{N}(A, \epsilon)$ becomes exponentially small compared to the contribution from $\blacksquare$, as $\epsilon \rightarrow 0$.

We now give you a wonderful theorem that provides the fractal dimension of the attractor of an important class of IFS. It will allow you to estimate fractal dimensions “on the fly,” simply from inspection of pictures of fractals, once you get used to it.

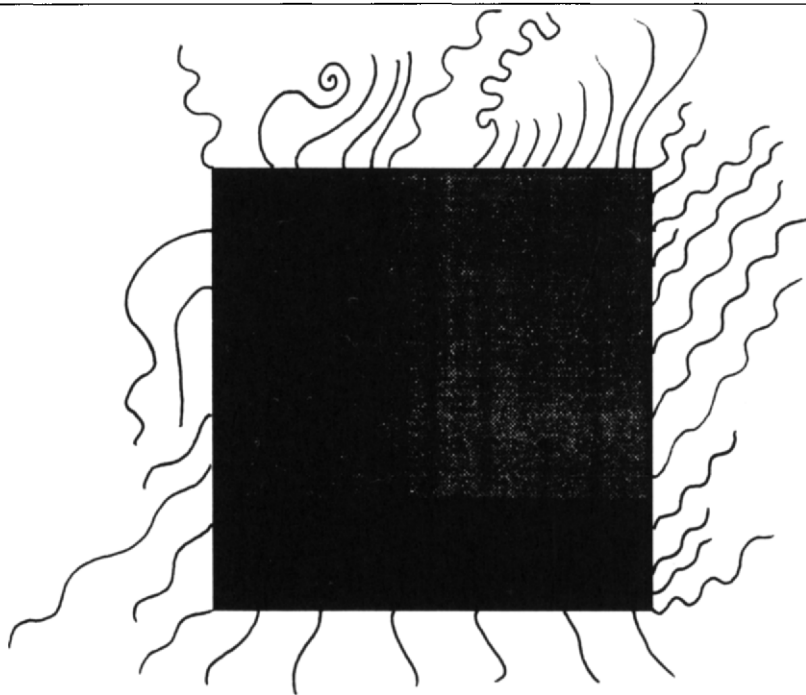


Figure V.138. Picture of a hairy box. The fractal dimension of the subset of $\mathbb{R}^2$ suggested here is the same as the fractal dimension of the box. The hairs are overpowered.

Theorem 2.3 Let $\{\mathbb{R}^m; w_1, w_2, \dots, w_N\}$ be a hyperbolic IFS, and let A denote its attractor. Suppose w_n is a similitude of scaling factor s_n for each $n \in \{1, 2, 3, \dots, N\}$. If the IFS is totally disconnected or just-touching then the attractor has fractal dimension $D(A)$, which is given by the unique solution of

$$\sum_{n=1}^N |s_n|^{D(A)} = 1, \quad D(A) \in [0, m].$$

If the IFS is overlapping, then $\bar{D} \geq D(A)$, where $\bar{D}$ is the solution of

$$\sum_{n=1}^N |s_n|^{\bar{D}} = 1, \quad \bar{D} \in [0, \infty).$$

Sketch of proof The full proof can be found in [Bedford 1986], [Hardin 1985], [Hutchinson 1981], and [Reuter 1987]. The following argument gives a valuable insight into the fractal dimension. We restrict attention to the case where the IFS $\{\mathbb{R}^m; w_1, w_2, \dots, w_N\}$ is totally disconnected. We suppose that the scaling factor s_i associated with the similitude w_i is nonzero for each $i \in \{1, 2, \dots, N\}$. Let $\epsilon > 0$. We begin by making two observations.

Observation (i): Let $i \in \{1, 2, \dots, N\}$. Since w_i is a similitude of scaling factor s_i , it maps closed balls onto closed balls, according to

$$w_i(B(x, \epsilon)) = B(w_i(x), |s_i|\epsilon).$$

Assume that $s_i \neq 0$. Then w_i is invertible, and obtain

$$w_i^{-1}(B(x, \epsilon)) = B(w_i^{-1}(x), |s_i|^{-1}\epsilon).$$

The latter two relations allow us to establish that for all $\epsilon > 0$,

$$\mathcal{N}(A, \epsilon) = \mathcal{N}(w_i(A), |s_i|\epsilon);$$

which is equivalent to

$$\mathcal{N}(w_i(A), \epsilon) = \mathcal{N}(A, |s_i|^{-1}\epsilon). \quad (1)$$

This applies for each $i \in \{1, 2, 3, \dots, N\}$.

Observation (ii): The attractor A of the IFS is the disjoint union

$$A = w_1(A) \cup w_2(A) \cup \dots \cup w_N(A),$$

where each of the sets $w_n(A)$ is compact. Hence we can choose the positive number ϵ so small that if, for some point $x \in \mathbb{R}^2$ and some integer $i \in \{1, 2, \dots, N\}$, we have $B(x, \epsilon) \cap w_i(A) \neq \emptyset$, then $B(x, \epsilon) \cap w_j(A) = \emptyset$ for all $j \in \{1, 2, \dots, N\}$ with $j \neq i$. It follows that if the number ϵ is sufficiently small we have

$$\mathcal{N}(A, \epsilon) = \mathcal{N}(w_1(A), \epsilon) + \mathcal{N}(w_2(A), \epsilon) + \mathcal{N}(w_3(A), \epsilon) + \dots + \mathcal{N}(w_N(A), \epsilon)$$

We put our two observations together. Substitute from equation 1 into the last equation to obtain

$$\begin{aligned} \mathcal{N}(A, \epsilon) &= \mathcal{N}(A, |s_1|^{-1}\epsilon) + \mathcal{N}(A, |s_2|^{-1}\epsilon) \\ &\quad + \mathcal{N}(A, |s_3|^{-1}\epsilon) + \dots + \mathcal{N}(A, |s_N|^{-1}\epsilon). \end{aligned} \quad (2)$$

This functional equation is true for all positive numbers ϵ that are sufficiently small. The proof is completed by showing formally that this implies the assertion in the theorem.

Here we demonstrate the reasonableness of the last step. Let us make the assumption $\mathcal{N}(A, \epsilon) \sim C\epsilon^{-D}$. Then substituting into equation 2 we obtain the equation:

$$C\epsilon^{-D} \approx C|s_1|^D\epsilon^{-D} + C|s_2|^D\epsilon^{-D} + C|s_3|^D\epsilon^{-D} + \dots + C|s_N|^D\epsilon^{-D}.$$

From this we deduce that

$$1 = |s_1|^D + |s_2|^D + |s_3|^D + \dots + |s_N|^D.$$

This completes our sketch of the proof of Theorem 2.3.

Examples & Exercises

2.2. This example takes place in the metric space $(\mathbb{R}^2, \text{Euclidean})$. A Sierpinski triangle is the attractor of a just-touching IFS of three similitudes, each with scaling factor 0.5. Hence the fractal dimension is the solution D of the equation

$$(0.5)^D + (0.5)^D + (0.5)^D = 1$$

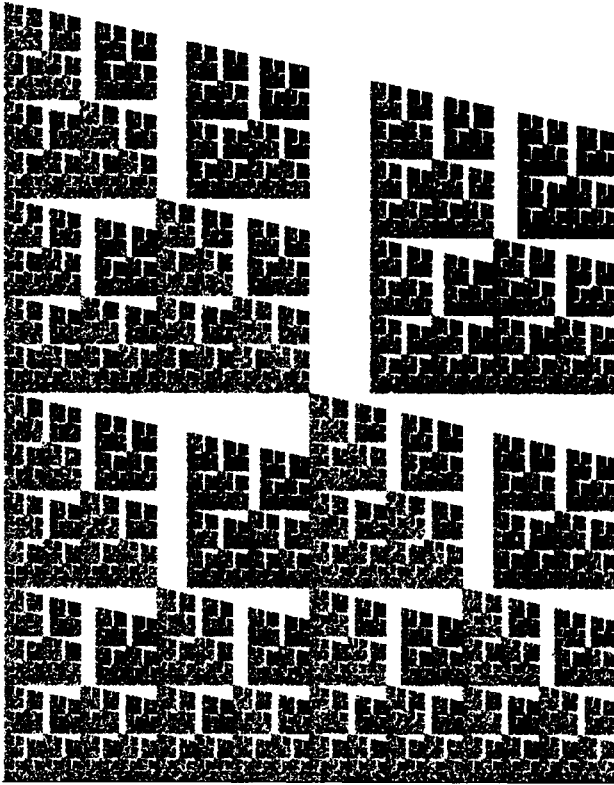


Figure V.139. The Castle fractal. This is an example of a self-similar fractal, and its fractal dimension may be calculated with the aid of Theorem 2.3. The associated IFS code is given in Table V.1.

from which we find

$$D = \frac{\ln(1/3)}{\ln(0.5)} = \frac{\ln(3)}{\ln(2)}.$$

2.3. Find a just-touching IFS of similitudes in $\mathbb{R}^2$ whose attractor is $\blacksquare$. Verify that Theorem 2.3 yields the correct value for the fractal dimension of $\blacksquare$.

2.4. The classical Cantor set is the attractor of the hyperbolic IFS

$$\{[0, 1]; w_1(x) = \frac{1}{3}x; w_2(x) = \frac{1}{3}x + \frac{2}{3}\}.$$

Use Theorem 2.3 to calculate its fractal dimension.

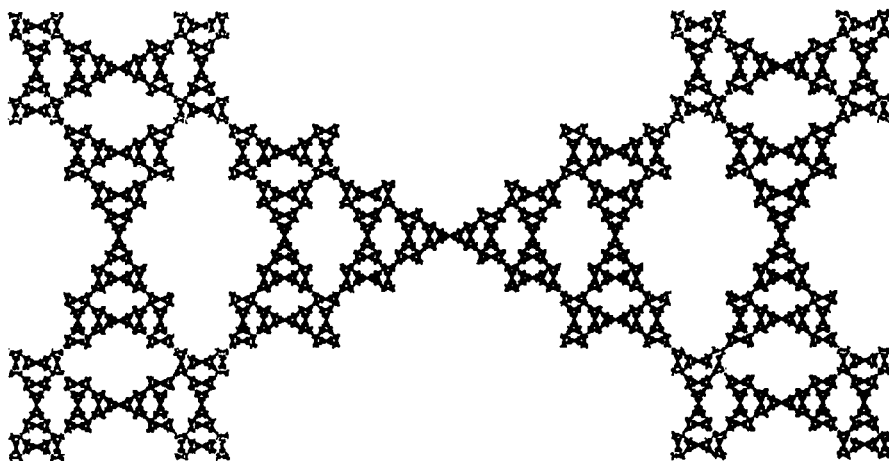
2.5. The attractor of a just-touching hyperbolic IFS $\{\mathbb{R}^2; w_i(x), i = 1, 2, 3, 4\}$ is represented in Figure V.139. The affine transformations $w_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are similitudes and are given in tabular form in Table V.1. Use Theorem 2.3 to calculate the fractal dimension of the attractor.

2.6. The attractor of a just-touching hyperbolic IFS $\{\mathbb{R}^2; w_i(x), i = 1, 2, 3\}$ is represented in Figure V.140. The affine transformations $w_i : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are similitudes.

Table V.1. IFS code for a Castle.

w	a	b	c	d	e	f	p
1	0.5	0	0	0.5	0	0	0.25
2	0.5	0	0	0.5	2	0	0.25
3	0.4	0	0	0.4	0	1	0.25
4	0.5	0	0	0.5	2	1	0.25

Figure V.140. To calculate the fractal dimension of the subset of $\mathbb{R}^2$ represented here, first apply the Collage Theorem to find a corresponding set of similitudes. Then use Theorem 2.3.



Use the Collage Theorem to find the similitudes, and then use Theorem 2.3 to calculate the fractal dimension of the attractor.

2.7. Figure V.141 represents the attractor of an overlapping hyperbolic IFS

$$\{\mathbb{R}^2; w_i(x), i = 1, 2, 3, 4\}.$$

Use the Collage Theorem and Theorem 2.3 to obtain an upper bound to the fractal dimension of the attractor.

2.8. Calculate the fractal dimension of the subset of $\mathbb{R}^2$ represented by Figure V.142.

2.9. Consider the attractor A of a totally disconnected hyperbolic IFS

$$\{\mathbb{R}^7; w_i(x), i = 1, 2\}$$

where the two maps

$$w_1 : \mathbb{R}^7 \rightarrow \mathbb{R}^7 \quad \text{and} \quad w_2 : \mathbb{R}^7 \rightarrow \mathbb{R}^7$$

are similitudes, of scaling factors s_1 and s_2 , respectively. Show that A is also the attractor of the totally disconnected hyperbolic IFS $\{\mathbb{R}^7; v_i(x), i = 1, 2, 3, 4\}$ where $v_1 = w_1 \circ w_1$, $v_2 = w_1 \circ w_2$, $v_3 = w_2 \circ w_1$, and $v_4 = w_2 \circ w_2$. Show that $v_i(x)$ is a similitude, and find its scaling factor, for $i = 1, 2, 3, 4$. Now apply Theorem 2.3 to

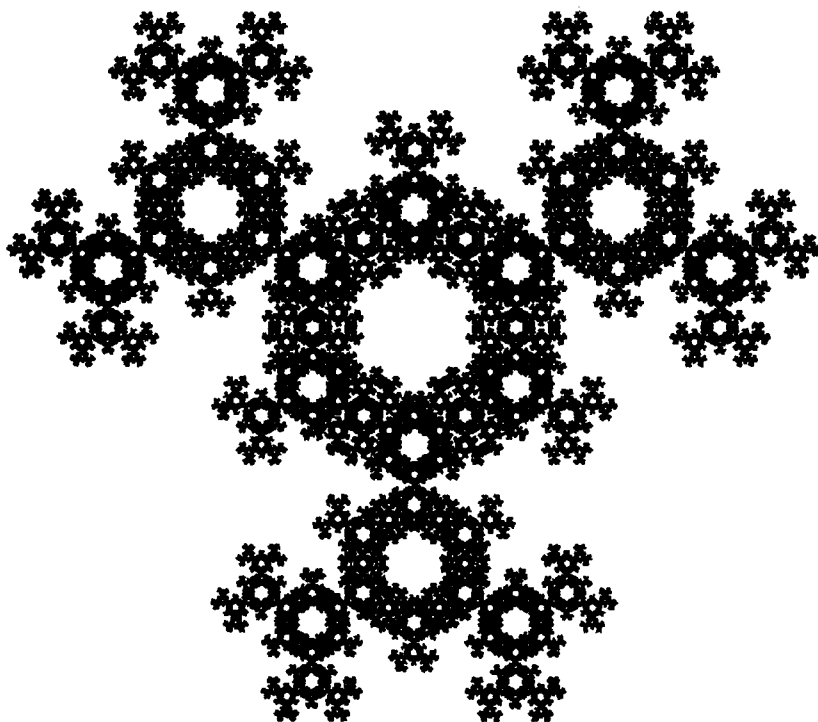


Figure V.141. An upper bound to the fractal dimension of the attractor of an overlapping IFS, corresponding to this picture, can be computed with the aid of Theorem 2.3.

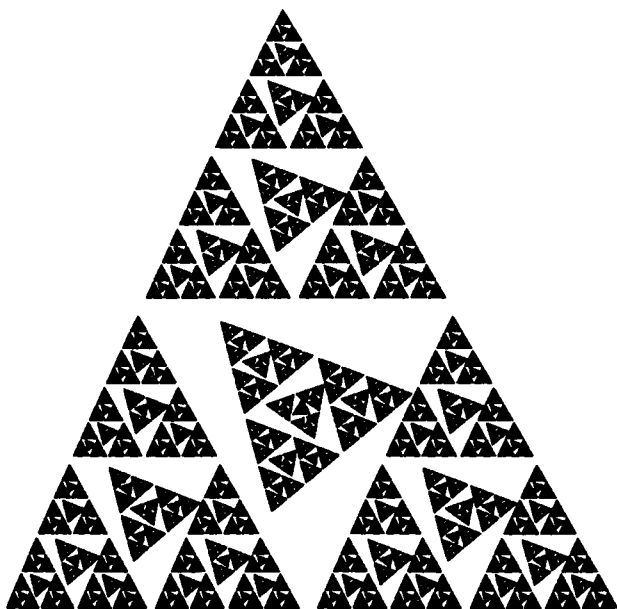


Figure V.142. Calculate the fractal dimension of the subset of $\mathbb{R}^2$ represented by this image.

yield two apparently different equations for the fractal dimension of A . Prove that these two equations have the same solution.

3 The Experimental Determination of the Fractal Dimension

In this section we consider the experimental determination of the fractal dimension of sets in the physical world. We model them, as best we can, as subsets of $(\mathbb{R}^2, \text{Euclidean})$ or $(\mathbb{R}^3, \text{Euclidean})$. Then, based on the definition of the fractal dimension, and sometimes in addition on one or another of the preceding theorems, such as the Box Counting Theorem, we analyze the model to provide a fractal dimension for the real-world set.

In the following examples we emphasize that when the fractal dimension of a physical set is quoted, some indication of how it was calculated must also be provided. There is not yet a broadly accepted unique way of associating a fractal dimension with a set of experimental data.

Example

3.1. There is a curious cloud of dots in the woodcut in Figure V.143. Let us try to estimate its fractal dimension by direct appeal to Definition 1.1.

We begin by covering the cloud of points by disks of radius ϵ for a range of ϵ -values from $\epsilon = 3$ cm down to $\epsilon = 0.3$ cm; and in each case we count the number of

Figure V.143. Covering a cloud of dots in a woodcut by balls of radius $\epsilon > 0$.

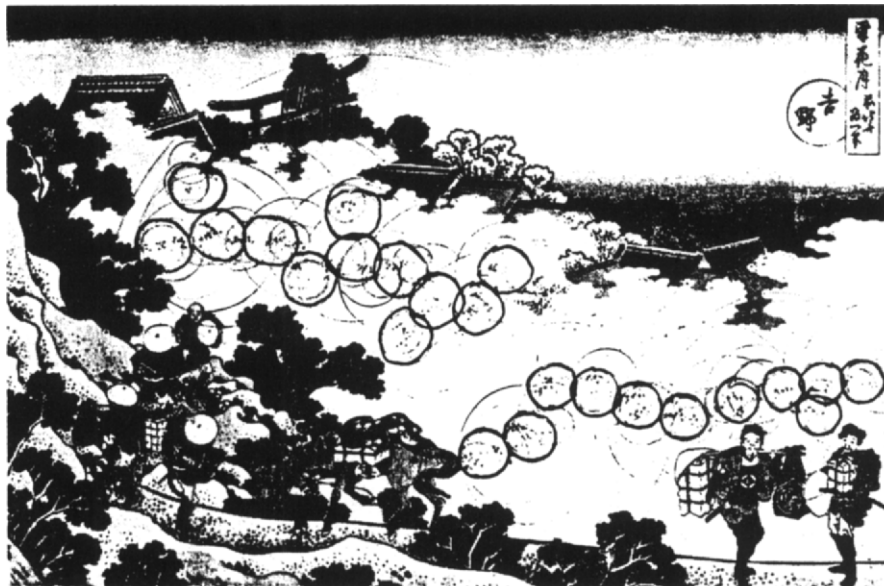


Table V.2. Minimal numbers of balls, of various radii, needed to cover a “dust” in a woodcut.

ϵ	$\mathcal{N}(A, \epsilon)$
3 cm	2
2 cm	3
1.5 cm	4
1.2 cm	6
1 cm	7
0.75 cm	10
0.5 cm	16
0.4 cm	23
0.3 cm	31
0.015 cm	267

Table V.3. The data in Table III.1 is tabulated in log–log form. These values are used to obtain the fractal dimension.

$\ln(1/\epsilon)$	$\ln(\mathcal{N}(A, \epsilon))$
-1.1	0.69
-0.69	1.09
-0.405	1.39
-0.182	1.79
0	1.95
0.29	2.30
0.693	2.77
0.916	3.13
1.204	3.43
4.2	5.59

disks needed. This provides the set of approximate values for $\mathcal{N}(A, \epsilon)$ given in Table III.1. The data is redisplayed in log–log format in Table V.3. The data in Table V.3 is plotted in Figure V.144. A straight line that approximately passes through the points is drawn. The slope of this straight line is our approximation to the fractal dimension of the cloud of points.

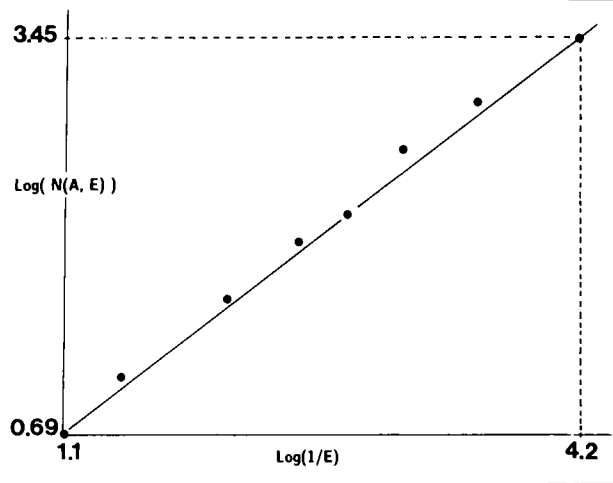
The experimental number $\mathcal{N}(A, 0.015 \text{ cm})$ is not very accurate. It is a very rough estimate based on the size of the dots themselves and is not included in the plot in Figure V.144. The slope of the straight line in Figure V.144 gives

$$D(A) \simeq 1.2, \text{ over the range } 0.3 \text{ cm to } 3 \text{ cm}, \quad (1)$$

where A denotes the set of points whose dimension we are approximating.

The straight line in Figure V.144 was drawn “by eye.” Thus if one was to repeat

Figure V.144. Log-log plot to estimate the fractal dimension D for the cloud of dots in the woodcut in Figure V.143. The data is in Tables III.1 and V.3.



the experiment, a different value for $D(A)$ may be obtained. In order to make the results consistent from experiment to experiment, the straight line should be estimated by a least squares method.

In proceeding by direct appeal to Definition 1.1, the estimates of $\mathcal{N}(A, \epsilon)$ need to be made very carefully. One needs to be quite sure that $\mathcal{N}(A, \epsilon)$ is indeed the *least* number of balls of radius ϵ needed. For large sets of data this could be very time-consuming.

It is clearly important to state the range of scales used: we have no idea or definition concerning the structure of the dots in Figure V.143 at higher resolutions than say 0.015 cm. Moreover, regardless of how much experimental data we have, and regardless of how many scales of observation are available to us, we will always end up estimating the slope of a straight line corresponding to a finite range of scales. If we include the data point (0.015 cm, 267) in the above estimation, we obtain

$$D(A) \simeq 0.9, \text{ over the range of scales } 0.015 \text{ to } 5 \text{ cm.} \quad (2)$$

We comment on the difference between the estimates in equations 1 and 2. If we restrict ourselves to the range of scales in equation 1, there little information present in the data to distinguish the cloud of points from a very irregular curve. However, the data used to obtain equation 2 contains values for $\mathcal{N}(A, \epsilon)$ for several values of ϵ such that the corresponding coverings of A are disconnected. The data is “aware” that A is disconnected. This lowers the experimentally determined value of D .

3.2. In this example we consider the physical set labelled A in Figure V.145. A is actually an approximation to a classical Cantor set. In this case we make an experimental estimate of the fractal dimension, based on the Box Counting Theorem. A Cartesian coordinate system is set up as shown and we attempt to count the number of square boxes $\mathcal{N}_n(A)$ of side $(1/2^n)$ which intersect A . We are able to obtain fairly accurate values of $\mathcal{N}_n(A)$ for $n = 0, 1, 2, 3, 4, 5$, and 6. These values are pre-

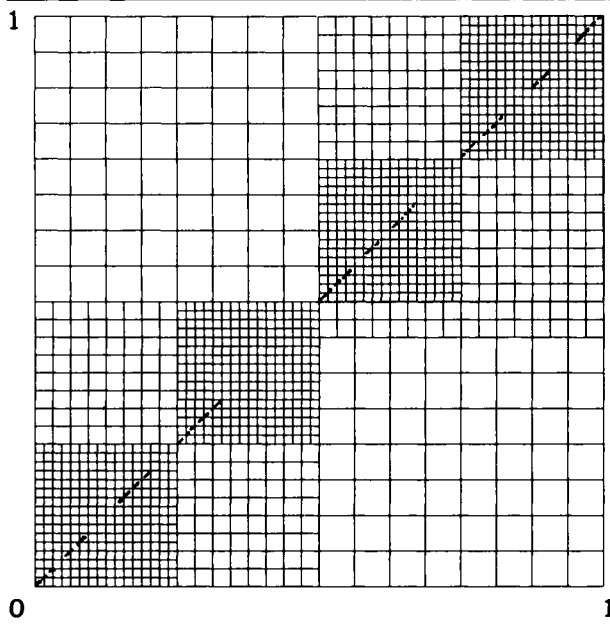


Figure V.145. Successive subdivision of overlaying grid to obtain the box counts needed for the application of Theorem 1.2 to estimate the fractal dimension of the Cantor set A . The counts are presented in Table V.4.

Table V.4. The data determined from Figure 145, in the experimental calculation of the fractal dimension of the physical set A .

n	$\mathcal{N}_n(A)$	$\ln \mathcal{N}_n(A)$	$n \ln 2$
0	1	0	0
1	3	1.10	0.69
2	7	1.95	1.38
3	10	2.30	2.08
4	19	2.94	2.77
5	33	3.50	3.46
6	58	4.06	4.16

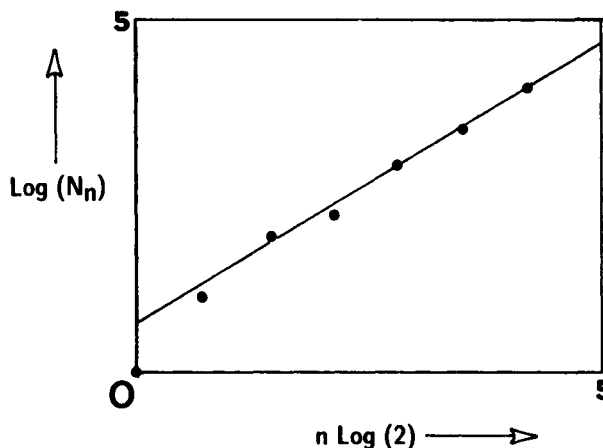
sented in Table V.4. We note that these values depend on the choice of coordinate system. Nonetheless the values of $\mathcal{N}_n(A)$ are much easier to measure than the values of $\mathcal{N}(A, \epsilon)$ used in example 3.1.

The analysis of the data proceeds just as in example 3.1. It is represented in Table V.4 and Figure V.146. We obtain

$$D(A) \simeq 0.8, \text{ over the range } \frac{1}{8} \text{ inch to 8 inches.}$$

3.3. In this example we show how a good experimentalist [Strahle 1987] overcomes the inherent difficulties with the experimental determination of fractal dimensions. In so doing he obtains a major scientific result. The idea is to compare two sets of experimental data, obtained by different means, on the same physical system. The

Figure V.146. Slope of the plot of the data in Table V.4 gives an approximation to the fractal dimension of the set A in Figure V.145.



physical system is a laboratory jet flame. The data are time series for the temperature and velocity at two different points in the jet. The idea is to apply the same procedure to the analysis of the two sets of data to obtain a value for the fractal dimension. The two values are the same. Instead of drawing the conclusion that the two sets of data “have the same fractal dimension,” he deduces that the two sets of data have a common source. That common source is physical, real-world chaos.

The experimental setup is as follows. A flame is probed by (a) a laser beam and (b) a very thin wire. These two probes, coupled with appropriate measuring devices, allow measurements to be made of the temperature and velocity in the jet, at two different points, as a function of time. In (a) the light bounces off the fast-moving molecules in the exhaust, and a receiver measures the characteristics of the bounced light. The output from the receiver is a voltage. This voltage, suitably rescaled, gives the temperature of the jet as a function of time. In (b) a constant temperature is maintained through a wire in the flame. The voltage required to hold the temperature constant is recorded. This voltage, suitably rescaled, gives the velocity of the jet as a function of time. In this way we obtain two independent readings of two different, but related, quantities.

Of course the experimental apparatus is much more sophisticated than it sounds from the above description. What is important is that the measuring devices are of very high resolution, accuracy, and sensitivity. A reading of the velocity can be made once every microsecond. In this example the temperature was read every 0.5×10^{-4} sec. Vast amounts of data can be obtained. A sample of the experimental output from (a) is shown in Figure V.147, where it is represented as the graph of voltage against time. It is a very complex curve. If one “magnifies up” the curve, one finds that its geometrical complexity in the curve continues to be present. It is just the sort of thing we fractal geometers like to analyze.

A sample of the experimental output from (b) is shown in Figure V.148, again represented as a graph of voltage against time. You should compare Figures V.147

RAYLEIGH SCATTERING VOLTAGE

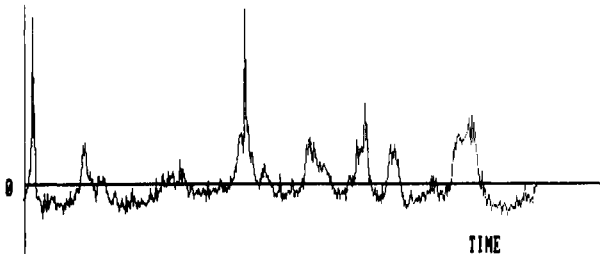


Figure V.147. Graph of voltage as a function of time from an experimental probe of a turbulent jet. In this case the probe measures scattering of a laser beam by the flame.

HOT FILM VOLTAGE

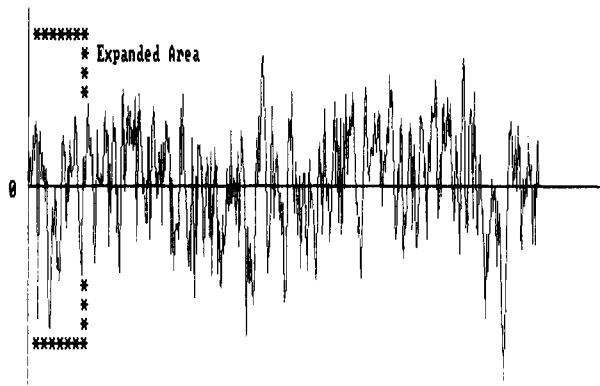


Figure V.148. Graph of voltage as a function of time from an experimental probe of a turbulent jet. In this case the probe measures the voltage across a wire in the flame. This data has a definite fractal character, as demonstrated by the expanded piece shown in Figure V.149.

and V.148. They look different. Is there a relationship between them? There should be: they both probe the same burning gas and they are in the same units.

In order to bring out the fractal character in the data, an expanded piece of the data in Figure V.148 is shown in Figure V.149.

The fractal dimensions of the graphs of the two time series, obtained from (a) and (b), is calculated using a method based on the Box Counting Theorem. Exactly the same method is applied to both sets of data, over the same range of scales. Figure V.150 shows the graphical analysis of the resulting box counts. Both experiments yield the same value

$$D \approx 1.5, \text{ over the range of scales } 2^6 \times 10^{-5} \text{ sec to } 2^{13} \times 10^{-5} \text{ sec.}$$

This suggests that, despite the different appearances of their graphs, there is a common source for the data.

We believe that this common source is chaotic dynamics of a certain special flavor and character, present in the jet flame. If so then fractal dimension provides an

experimentally measurable parameter that can be used to characterize the brand of chaos.

3.4. Use a method based on the direct application of Definition 1.1 to make an experimental determination of the fractal dimension of the physical set defined by the black ink in Figure V.151. Give the range of scales to which your result applies.

3.5. Use a method based on the Box Counting Theorem, as in Example 3.2, to estimate the fractal dimension of the “random dendrite” given in Figure V.152. State the range of scales over which your estimate applies. Make several complete experiments to obtain some idea of the accuracy of your result.

3.6. Make an experimental estimate of the fractal dimension of the dendrite shown in Figure V.153. Note that a grid of boxes of size $(1/12)$ th inch by $(1/12)$ th inch has been printed on top of the dendrite. Compare the result you obtain with the result of exercise 3.5. It is important that you follow exactly the same procedure in both experiments.

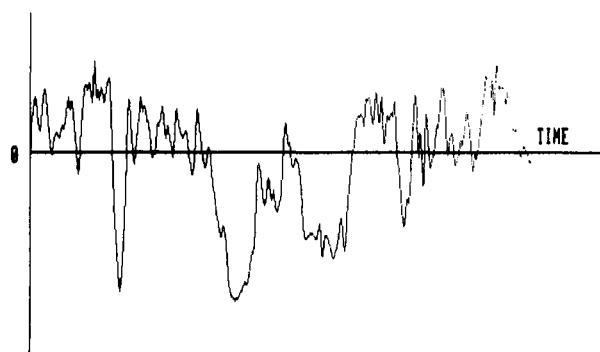
3.7. Make an experimental determination of the fractal dimension of the set in Figure V.142. Compare your result with a theoretical estimate based on Theorem 2.3, as in exercise 2.7.

3.8. Obtain maps of Great Britain of various sizes. Make an experimental determination of the fractal dimension of the coastline, over as wide a range of scales as possible.

3.9. Obtain data showing the variations of a Stock Market index, at several different time scales, for example, hourly, daily, monthly, and yearly. Make an experimental determination of the fractal dimension. Find a second economic indicator for the same system and analyze its fractal dimension. Compare the results.

Figure V.149. A blowup of a piece of the graph in Figure V.148.

EXPANDED AREA HOT FILM VOLTAGE



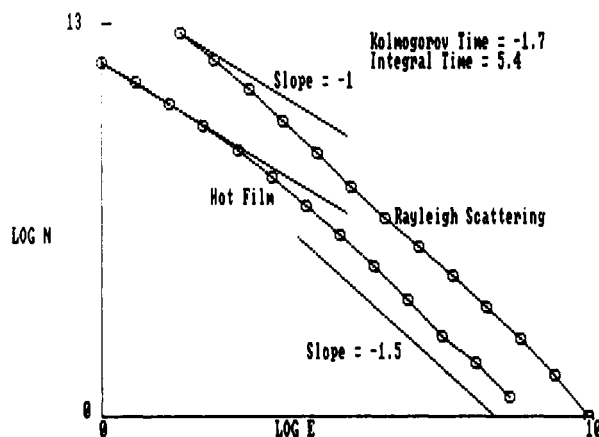


Figure V.150. Graphical analysis of box counts associated with experiments (a) and (b). The data analyzed is illustrated in Figures V.145 and V.148. The two data streams, which come from probes of a single turbulent system, when analyzed in exactly the same way, yield the same value $D = 1.5$ for the fractal dimension. This suggests that, despite the different appearances of their graphs, there is a common source for the data. This source is chaotic dynamics of a certain special flavor and character. The fractal dimension provides a measurable symptom of the brand of chaos.

4 The Hausdorff-Besicovitch Fractal Dimension

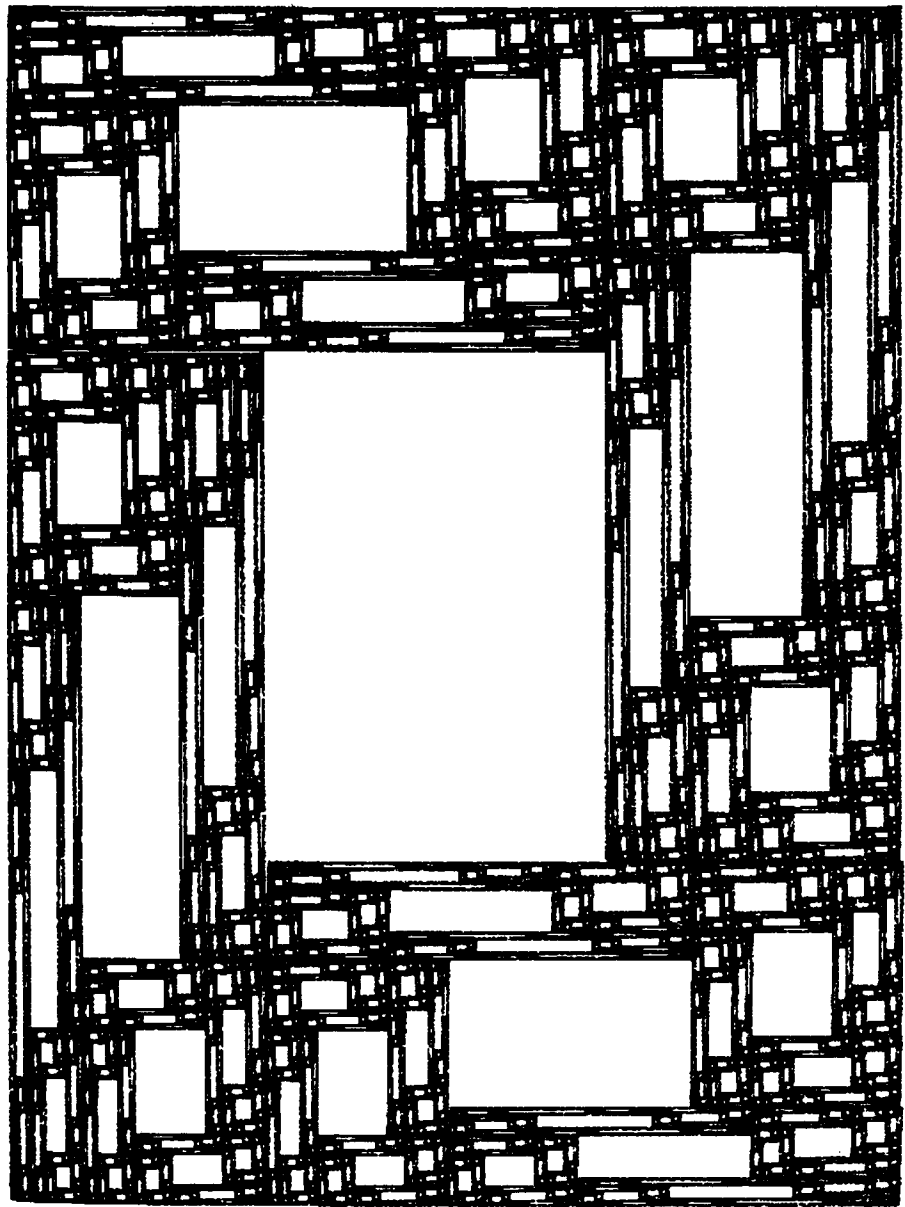
The Hausdorff-Besicovitch dimension of bounded subset of $\mathbb{R}^m$ is another real number that can be used to characterize the geometrical complexity of bounded subsets of $\mathbb{R}^m$. Its definition is more complex and subtle than that of the fractal dimension. One of the reasons for its importance is that it is associated with a method for comparing the “sizes” of sets whose fractal dimensions are the same. It is harder to work with than the fractal dimension, and its definition is not usually used as the basis of experimental procedures for the determination of fractal dimensions of physical sets.

Throughout we work in the metric space $(\mathbb{R}^m, d)$ where m is a positive integer and d denotes the Euclidean metric. Let $A \subset \mathbb{R}^m$ be bounded. Then we use the notation

$$\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}.$$

Let $0 < \epsilon < \infty$, and $0 \leq p < \infty$. Let $\mathcal{A}$ denote the set of sequences of subsets $\{A_i \subset A\}$, such that $A = \bigcup_{i=1}^{\infty} A_i$. Then we define

Figure V.151. Make an experimental estimate of the fractal dimension of the set A of black ink, above, over the range of scales 5 inches to 0.1 inches. Base your experimental method directly on Definition 1.1.



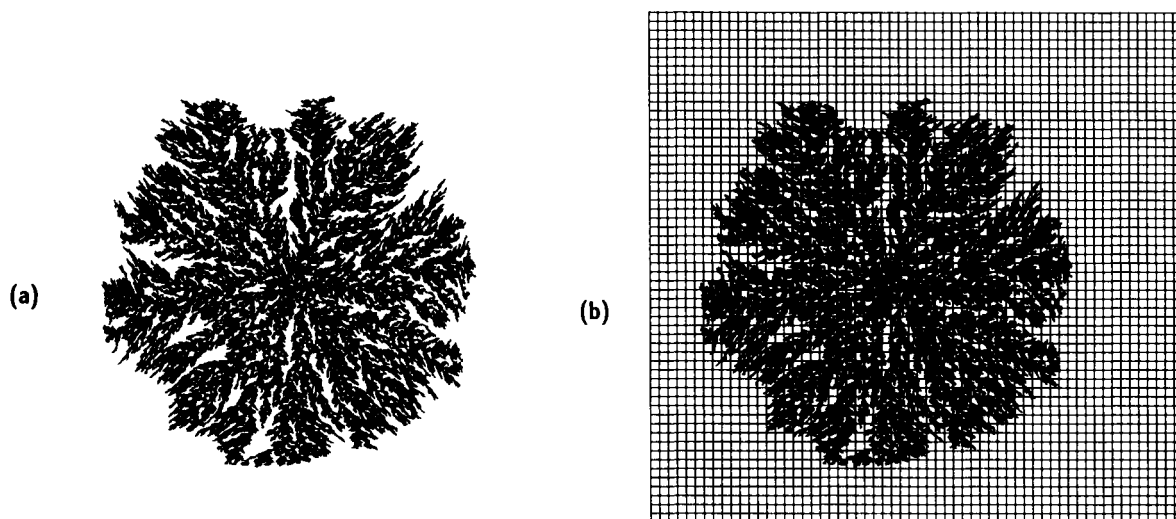


Figure V.152. Make an experimental estimate of the fractal dimension of this set in (a), over the range of scales 5 inches to $(1/12)$ th inch, basing your method on the Box Counting Theorem and graphical analysis. In order to help you with your work, in (b) we have overlaid the set a grid of boxes $(1/12)$ th inch by $(1/12)$ th inch.

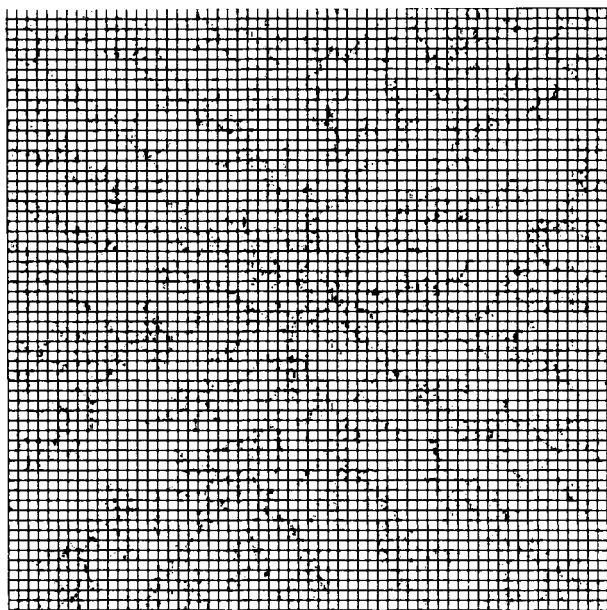


Figure V.153. Make an experimental estimate of the fractal dimension of the random dendrite shown here. Note that a grid of boxes of size $(1/12)$ th inch by $(1/12)$ th inch has been printed on top of the dendrite. Compare the experimental fractal dimension here with that of the dendrite in Figure V.152. In advance, which one do you expect will have lower fractal dimension?

$$\mathcal{M}(A, p, \epsilon) = \inf \left\{ \sum_{i=1}^{\infty} (\text{diam}(A_i))^p : \{A_i\} \in \mathcal{A}, \text{ and } \text{diam}(A_i) < \epsilon \quad \text{for } i = 1, 2, 3, \dots \right\}.$$

Here we use the convention that $(\text{diam}(A_i))^0 = 0$ when A_i is empty. $\mathcal{M}(A, p, \epsilon)$ is a number in the range $[0, \infty]$; its value may be zero, finite, or infinite. You should verify that it is a nonincreasing function of ϵ . We now define

$$\mathcal{M}(A, p) = \sup\{\mathcal{M}(A, p, \epsilon) : \epsilon > 0\}.$$

Then for each $p \in [0, \infty]$ we have $\mathcal{M}(A, p) \in [0, \infty]$.

Definition 4.1 Let m be a positive integer and let A be a bounded subset of the metric space $(\mathbb{R}^m, \text{Euclidean})$. For each $p \in [0, \infty)$ the quantity $\mathcal{M}(A, p)$ described above is called the Hausdorff p -dimensional measure of A .

Examples & Exercises

- 4.1.** Show that $\mathcal{M}(A, p)$ is a nonincreasing function of $p \in [0, \infty]$.
- 4.2.** Let A denote a set of seven distinct points in $(\mathbb{R}^2, \text{Euclidean})$. Show that $\mathcal{M}(A, 0) = 7$ and $\mathcal{M}(A, p) = 0$ for $p > 0$.
- 4.3.** Let A denote a countable infinite set of distinct points in $(\mathbb{R}^2, \text{Euclidean})$. Show that $\mathcal{M}(A, 0) = \infty$ and $\mathcal{M}(A, p) = 0$ for $p > 0$.
- 4.4.** Let $\mathcal{C}$ denote the classical Cantor set in $[0, 1]$. Show that $\mathcal{M}(\mathcal{C}, 0) = \infty$ and $\mathcal{M}(\mathcal{C}, 1) = 0$.
- 4.5.** Let Δ denote a convenient Sierpinski triangle. Show that $\mathcal{M}(\Delta, 1) = \infty$ and $\mathcal{M}(\Delta, 2) = 0$. Can you evaluate $\mathcal{M}(\Delta, \ln(3)/\ln(2))$? At least try to argue why this might be an interesting number.

The Hausdorff p -dimensional measure $\mathcal{M}(A, p)$, as a function of $p \in [0, \infty]$, behaves in a remarkable manner. Its range consists of only one, two, or three values! The possible values are zero, a finite number, and infinity. In Figure V.154 we illustrate this behavior when A is a certain Sierpinski triangle.

Theorem 4.1 Let m be a positive integer. Let A be a bounded subset of the metric space $(\mathbb{R}^m, \text{Euclidean})$. Let $\mathcal{M}(A, p)$ denote the function of $p \in [0, \infty)$ defined above. Then there is a unique real number $D_H \in [0, m]$ such that

$$\mathcal{M}(A, p) = \begin{cases} \infty & \text{if } p < D_H \text{ and } p \in [0, \infty), \\ 0 & \text{if } p > D_H \text{ and } p \in [0, \infty). \end{cases}$$

Proof This can be found, for example, in [Federer 1969], section 2.10.3.

Definition 4.2 Let m be a positive integer and let A be a bounded subset of the metric space $(\mathbb{R}^m, \text{Euclidean})$. The corresponding real number D_H , occurring in Theorem 4.1, is called the Hausdorff-Besicovitch dimension of the set A . This number will also be denoted by $D_H(A)$.

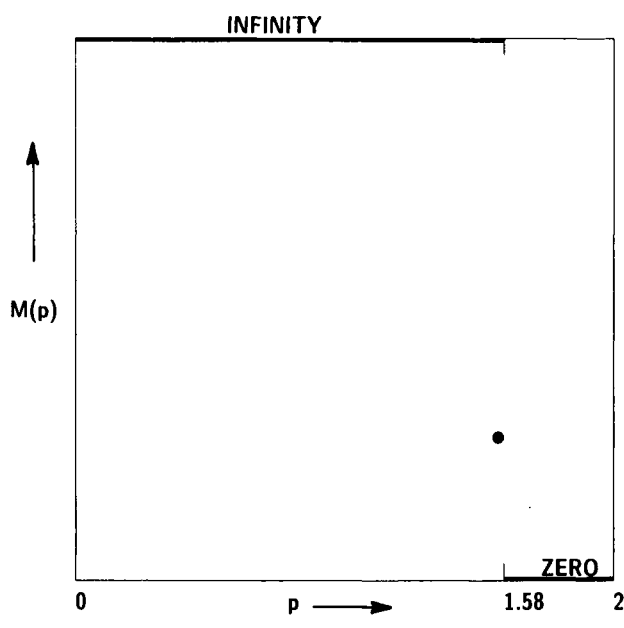


Figure V.154. Graph the function $\mathcal{M}(\Delta, p)$ when Δ is a certain Sierpinski triangle. It takes only three values.

Theorem 4.2 Let m be a positive integer and let A be a subset of the metric space $(\mathbb{R}^m, \text{Euclidean})$. Let $D(A)$ denote the fractal dimension of A and let $D_H(A)$ denote the Hausdorff-Besicovitch dimension of A . Then

$$0 \leq D_H(A) \leq D(A) \leq m.$$

Examples & Exercises

4.6. Describe a situation where you would expect $D_H(A) < D(A)$.

4.7. Prove Theorem 4.2.

Theorem 4.3 Let m be a positive integer. Let $\{\mathbb{R}^m; w_1, w_2, \dots, w_N\}$ be a hyperbolic IFS, and let A denote its attractor. Let w_n be a similitude of scaling factor s_n for each $n \in \{1, 2, 3, \dots, N\}$. If the IFS is totally disconnected or just-touching, then the Hausdorff-Besicovitch dimension $D_H(A)$ and the fractal dimension $D(A)$ are equal. In fact $D(A) = D_H(A) = D$, where D is the unique solution of

$$\sum_{n=1}^N |s_n|^D = 1, \quad D \in [0, m].$$

If D is positive, then the Hausdorff D -dimensional measure $\mathcal{M}(A, D_H(A))$ is a positive real number.

Proof This can be found in [Hutchinson 1981].

In the situation referred to in Theorem 4.3 the Hausdorff $D_H(A)$ -dimensional measure can be used to compare the “sizes” of fractals that have the same fractional dimension. The larger the value of $\mathcal{M}(A, D_H(A))$, the “larger” the fractal. Of course, if two fractals have different fractal dimensions, then we say that the one with the higher fractal dimension is the “larger” one.

Examples & Exercises

4.8. Here we provide some intuition about the functions $\mathcal{M}(A, p, \epsilon)$ and $\mathcal{M}(A, p)$, and the “sizes” of fractals. We illustrate how these quantities can be estimated. The type of procedure we use can often be followed for attractors of just-touching and totally disconnected IFS whose maps are all similitudes and should lead to correct values. Formal justification is tedious and follows the lines suggested in [Hutchinson 1981].

Consider the Sierpinski triangle Δ with vertices at $(0, 0)$, $(0, 1)$, and $(1, 0)$. We work in $\mathbb{R}^2$ with the Euclidean metric. We begin by estimating the number $\mathcal{M}(\Delta, p, \epsilon)$ for $p \in [0, 1]$ for various values of ϵ . The values of ϵ we consider are $\epsilon = \sqrt{2}(1/2)^n$ for $n = 0, 1, 2, 3, \dots$. Now notice that Δ can be covered very efficiently by 3^n closed disks of radius $\sqrt{2}(1/2)^n$. We guess that this covering is one for which the infimum in the definition of $\mathcal{M}(\Delta, p, \epsilon = \sqrt{2}(1/2)^n)$ is actually achieved. We obtain the estimate

$$\mathcal{M}(\Delta, p, \sqrt{2}(1/2)^n) = 3^n (\sqrt{2})^p (1/2)^{np} \text{ for } n = 1, 2, 3, \dots$$

The supremum in the definition of $\mathcal{M}(\Delta, p)$ can be replaced by a limit; so we obtain

$$\begin{aligned} \mathcal{M}(\Delta, p) &= \lim_{n \rightarrow \infty} \{3^n (\sqrt{2})^p (1/2)^{np}\} \\ &= \begin{cases} \infty & \text{if } p < \ln(3)/\ln(2), \\ (\sqrt{2})^{\ln(2)/\ln(3)} & \text{if } p = \ln(3)/\ln(2), \\ 0 & \text{if } p > \ln(3)/\ln(2). \end{cases} \end{aligned}$$

This tells us that $D_H(\Delta) = \ln(3)/\ln(2)$, which we already know from Theorem 4.3. It also tells us that $\mathcal{M}(\Delta, D_H(\Delta)) = (\sqrt{2})^{\ln(2)/\ln(3)}$. This is our estimate of the “size” of the particular Sierpinski triangle under consideration.

If one repeats the above steps for the Sierpinski triangle $\tilde{\Delta}$ with vertices at $(0, 0)$, $(0, 1/\sqrt{2})$, and $(1/\sqrt{2}, 0)$, one finds $\mathcal{M}(\tilde{\Delta}, D_H(\tilde{\Delta})) = 1$. Thus $\tilde{\Delta}$ is “smaller” than Δ . Similar estimates can be made for pairs of attractors of totally disconnected or just-touching IFS whose maps are similitudes and whose fractal dimensions are equal. The comparison of “sizes” becomes exciting when the two attractors are not metrically equivalent.

4.9. Estimate the “sizes” of the two fractals represented in Figure V.155. Which one is “largest”? Does the computed estimate agree with your subjective feeling about which one is largest?

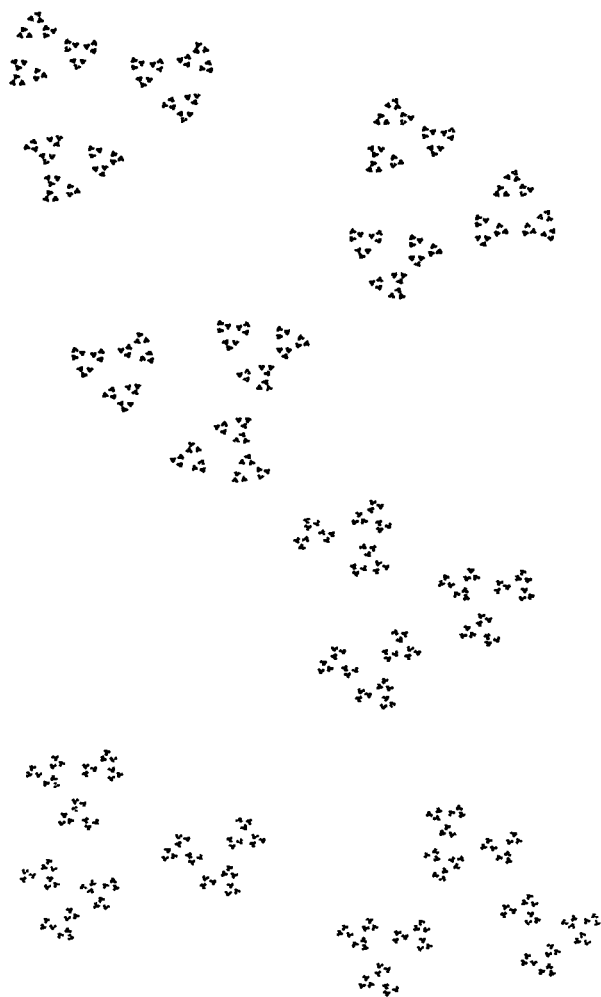


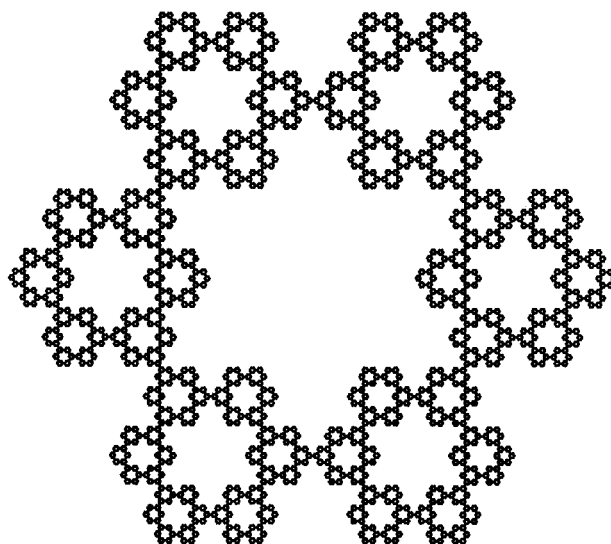
Figure V.155. The two images here represent the attractors of two different IFS of the form $\{\mathbb{R}^2; w_1, w_2, w_3\}$, where all of the maps are similitudes of scaling factor 0.4. Both sets have the same fractal dimension $\ln(3)/\ln(2.5)$. So which one is the “largest”? Compare their “sizes” by estimating their Hausdorff $\ln(3)/\ln(2.5)$ -dimensional measures.

4.10. Prove that the Hausdorff-Besicovitch dimension of two metrically equivalent bounded subsets of $(\mathbb{R}^m, \text{Euclidean})$ is the same.

4.11. Let d denote a metric on $\mathbb{R}^2$ which is equivalent to the Euclidean metric. Let A denote a bounded subset of $\mathbb{R}^2$. Suppose that d is used in place of the Euclidean metric to calculate a “Hausdorff-Besicovitch” dimension of A , denoted by $\tilde{D}_H(A)$. Prove that $D_H(A) = \tilde{D}_H(A)$. Show, however, that the “size” of the set, $\mathcal{M}(A, D_H(A))$, may be different when computed using d in place of the Euclidean metric.

4.12. If distance in $\mathbb{R}^2$ is measured in inches, and a subset A of $\mathbb{R}^2$ has fractal dimension 1.391, what are the units of $\mathcal{M}(A, 1.391)$?

Figure V.156. Why are the fractal dimension and the Hausdorff-Besicovitch dimension of the attractor of the IFS represented by this image equal?



4.13. The image in Figure V.156 represents the attractor A of a certain hyperbolic IFS.

- (1) Explain, with support from appropriate theorems, why the fractal dimension D and the Hausdorff-Besicovitch dimension D_H of the attractor of the IFS are equal.
- (2) Evaluate D .
- (3) Using inches as the unit, compare the Hausdorff-Besicovitch D -dimensional measures of A and $w(A)$, where $w(A)$ denotes one of the small “first-generation” copies of A .

4.14. By any means you like, estimate the Hausdorff-Besicovitch dimension of the coastline of Baron von Koch’s Island, shown in Figure V.157. It is recommended that theoreticians try to make an experimental estimate, and that experimentalists try to make a theoretical estimate.

4.15. Does the work of some artists have a characteristic fractal dimension? Make a comparison of the empirical fractal dimensions of Romeo and Juliet, over an appropriate range of scales; see Figure V.158.

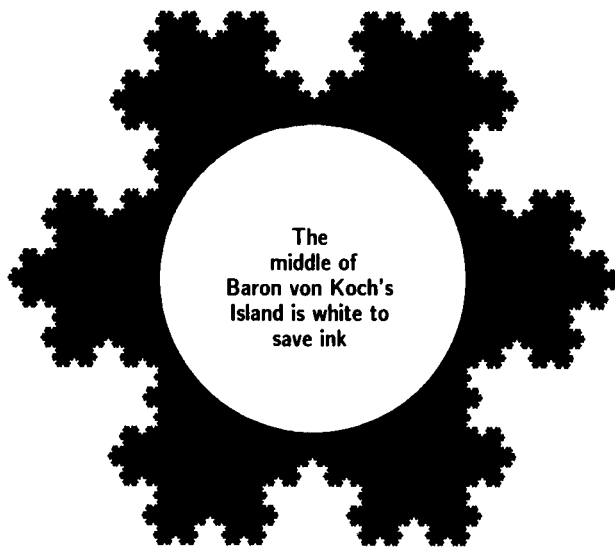


Figure V.157. By any means you like, estimate the Hausdorff-Besicovitch dimension of the coastline of Baron von Koch's Island.

Figure V.158. Does the work of some artists have a characteristic fractal dimension? Make a comparison of the empirical fractal dimensions of Romeo and Juliet, over an appropriate range of scales.

